

Question 1 (MCQ):

1. Which of the following is true about the for loop in Java?

- A) The initialization expression is evaluated once at the start.
- B) The condition expression is evaluated before every iteration.
- C) The update expression is evaluated after every iteration.
- D) All of the above

2. What will be the output of the following code?

```
int i = 0;
while (i < 5) {
    System.out.print(i + " ");
    i++;
    if (i == 3) break;
}
```

- A) 0 1 2
- B) 0 1 2 3 4
- C) 1 2 3
- D) 0 1

3. Consider the code snippet below. What will be the output?

```
for (int i = 0; i < 5; i++) {
    for (int j = 0; j < 3; j++) {
        System.out.print(i + j + " ");
    }
}
```

- A) 0 1 2 1 2 3 2 3 4 3 4 5 4 5 6
- B) 0 1 2 3 4 5 6 7 8
- C) 0 1 2 1 2 3 2 3 4 3 4 5 4 5 6
- D) 0 1 2 1 2 3 2 3 4 3 4 5 4 5 6 5 6 7

4. Given the following code, what will be the output?

```
int sum = 0;
for (int i = 1; i <= 5; i++) {
    if (i % 2 == 0) continue;
    sum += i;
}
System.out.println(sum);
```

- A) 9
- B) 6
- C) 15
- D) 10

5. Which of the following loop constructs does not support the use of a continue statement?

- A) for
- B) while
- C) do-while
- D) None of the above

6. What is the return type of the main method in Java?

- A) void
- B) int
- C) String
- D) None of the above

7. Which of the following is true about method overloading?

- A) Overloaded methods must have different return types.
- B) Overloaded methods must have different parameter lists.
- C) Overloaded methods must have different names.
- D) Overloaded methods can have the same parameter lists

8. Given the following code, what will be the output?

```
public class Main {
    public static void main(String[] args) {
        int result = sum(10, 20);
        System.out.println(result);
    }

    static int sum(int a, int b) {
        return a + b;
    }
}
```

9. What is the purpose of the return statement in a function?

- A) To terminate the function and transfer control back to the calling function.
- B) To pass control to the next iteration of a loop.
- C) To exit the program.
- D) To continue the execution of the function.

10. Which of the following is true about constructors in Java?

- A) Constructors can have any name.
- B) Constructors must have the same name as the class.
- C) Constructors can have a return type.
- D) Constructors cannot be overloaded.

11. What will be the output of the following code?

```
public class Main {  
    public Main() {  
        System.out.println("Constructor called");  
    }  
  
    public static void main(String[] args) {  
        Main obj = new Main();  
    }  
}
```

- A) No output
- B) Runtime error
- C) Constructor called
- D) Compilation error

12. Consider the following class definition. Which constructor will be called when new MyClass(10) is executed?

```
public class MyClass {  
    public MyClass() {  
        System.out.println("Default constructor");  
    }  
  
    public MyClass(int x) {  
        System.out.println("Parameterized constructor");  
    }  
}
```

- A) Default constructor
- B) Parameterized constructor
- C) No constructor
- D) Compilation error

13. What will be the output of the following code?

```
public class Main {  
    int x;  
  
    public Main() {  
        x = 5;  
    }  
  
    public static void main(String[] args) {  
        Main obj = new Main();  
        System.out.println(obj.x);  
    }  
}
```

- A) 0
- B) 5
- C) null
- D) Compilation error

14. Which of the following is not a valid way to create an object in Java?

- A) new ClassName();
- B) ClassName obj = new ClassName();
- C) ClassName obj = ClassName();
- D) new ClassName().methodName();

15. What will be the output of the following code?

```
public class Main {  
    public Main(int x) {  
        System.out.println(x);  
    }  
}
```

```

public static void main(String[] args) {
    Main obj = new Main(10);
}
}

```

A) 0
B) 10
C) Compilation error
D) Runtime error

16. What will be the output of the following code?

```

public class Main {
    public static void main(String[] args) {
        int a = 5;
        modify(a);
        System.out.println(a);
    }

    static void modify(int x) {
        x = 10;
    }
}

```

A) 5
B) 10
C) 0
D) Compilation error

17. Given the following code, what will be the output?

```

public class Main {
    public static void main(String[] args) {
        int[] arr = {1, 2, 3, 4, 5};
        Main ob=new Main();
        ob.modifyArray(arr);
        for (int i=0; i<arr.length; i++) {
            System.out.print(arr[i] + " ");
        }
    }

    void modifyArray(int[] arr) {
        arr[0] = 99;
    }
}

```

A) 1 2 3 4 5
B) 99 2 3 4 5
C) 0 2 3 4 5
D) Compilation error

18. Consider the following code snippet. What will be the output?

```

public class Main {
    public static void main(String[] args) {
        int result = calculate(2, 3);
        System.out.println(result);
    }

    static int calculate(int a, int b) {
        int sum = 0;
        for (int i = 0; i < b; i++) {
            sum += a;
        }
        return sum;
    }
}

```

A) 5
B) 6
C) 8
D) 9

19. What will be the output of the following code?

```

public class Main {
    public static void main(String[] args) {
        Data d = new Data();
        d.value = 5;
        modifyData(d);
        System.out.println(d.value);
    }
}

```

```
static void modifyData(Data obj) {
    obj.value = 10;
}
}
```

```
class Data {
    int value;
}
```

- A) 5
- B) 10
- C) Compilation error
- D) NullPointerException

20. What will be the output of the following code?

```
public class Main {
    public static void main(String[] args) {
        int[] arr = {1, 2, 3, 4, 5};
        for (int i = 0; i < arr.length; i++) {
            if (i % 2 == 0) {
                arr[i] *= 2;
            }
        }
        for (int i = 0; i < arr.length; i++) {
            System.out.print(arr[i] + " ");
        }
    }
}
```

- A) 1 2 3 4 5
- B) 2 2 6 4 10
- C) 1 4 3 8 5
- D) Compilation error

Question 2. Define a function `int fact(int n)` to return factorial of `n`. Define another function `int prime(int n)` to return 1 if `n` is prime otherwise 0.

Write the main function to take a number and print factorial of its prime digits using the above functions.

Question 3. Overload a function `calculate()` for the following and call as user chooses.

find `s` where `s = 1 + (1*2) + (1*2*3) + .... + n`

or print

```
*
***
*****
*****
```

Question 4:

An electronics shop has announced a special discount on the purchase of Laptops as given below:

Category	Discount on Laptop
Up to ₹25,000	5.0%
₹25,001 - ₹50,000	7.5%
₹50,001 - ₹1,00,000	10.0%
More than ₹1,00,000	15.0%

Define a class Laptop described as follows:

Data members/instance variables:

1. name
2. price
3. dis
4. amt

Member Methods:

1. A parameterised constructor to initialize the data members
2. To accept the details (name of the customer and the price)
3. To compute the discount
4. To display the name, discount and amount to be paid after discount.